

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 5: Mercator & Transverse Mercator Projections

Syllabus Source: CAE Oxford (pp. 167-206) | Pass Target: $\geq 90\%$ AviationExam

1. THE DIRECT MERCATOR PROJECTION ARCHITECTURE

The Direct Mercator is a cylindrical conformal projection where the cylinder is tangent to the globe along the Equator:

- Meridians & Parallels: Meridians are parallel, equidistant straight vertical lines. Parallels are straight horizontal lines whose spacing EXPANDS rapidly away from the Equator!
- Rhumb Lines (The Great Benefit): RHUMB LINES ARE STRAIGHT LINES! A straight line drawn between any two points represents a constant compass track. This made Mercator the supreme marine and navigation chart for centuries.
- Great Circles on Mercator: Great Circles are CURVED LINES, bowing towards the nearer pole (lying on the polar side of the rhumb line).
- Scale Variation: Scale is correct at the Equator ($SF = 1.0$), and expands with the secant of latitude: $\text{Scale at Latitude} = \text{Scale at Equator} \times \sec(\text{Latitude}) = \text{Scale at Equator} / \cos(\text{Latitude})!$

2. TRANSVERSE MERCATOR PROJECTION

The cylinder is rotated 90° so it is tangent along a SINGLE MERIDIAN (the Central Meridian):

- Usage: Ideal for North-South flight corridors, topographical national grid surveys (OSGB, UTM), and high-latitude strip charts within 3° of the Central Meridian.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Mercator Chart Scale at High Latitude

SCENARIO (Standard EASA Chart Scale Problem): A Direct Mercator chart has a nominal scale of 1:2,000,000 at the Equator: QUESTION: What is the actual chart scale at 60° North latitude? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Recall the Mercator Scale Expansion Formula: - $\text{Scale at Latitude} = \text{Scale at Equator} \times \sec(\text{Latitude})$ - Or in Representative Fraction (RF) denominator terms: - Denominator at Latitude = Denominator at Equator $\times \cos(\text{Latitude})!$ Step 2: Plug in the numbers for 60° Latitude: - Denominator at Equator = 2,000,000 - Latitude = $60^\circ\text{N} \rightarrow \cos(60^\circ) = 0.50$ exactly! - Denominator at $60^\circ\text{N} = 2,000,000 \times 0.50 = 1,000,000!$ Step 3: State the New Scale: - New Scale at $60^\circ\text{N} = 1 : 1,000,000!$ Step 4: Physical Meaning for Dummies: - The scale is twice as large (features appear twice as big on paper) at 60°N than at the Equator, because land masses are stretched by a factor of $1/\cos(60^\circ) = 2.0!$ FINAL ANSWER: The scale at 60°N is 1:1,000,000.

[!] EXAM TRAP (AVIATIONEXAM) - The Scale Fraction vs Denominator Trap

- When scale 'EXPANDS' (gets larger), the DENOMINATOR GETS SMALLER! - 1:1,000,000 is a LARGER scale than 1:2,000,000. - Do not multiply the denominator by $\sec(\text{lat})$; multiply the denominator by $\cos(\text{lat})!$

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: On a Direct Mercator chart, how do RHUMB LINES appear?

[A] Complex sine curves [B] STRAIGHT LINES [C] Circles centered on the pole [D] Parabolas

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The unique defining property of the Mercator projection is that any straight line drawn across it intersects all meridians at a constant angle, representing a rhumb line.

Q2: How does a GREAT CIRCLE track between two airports in the Northern Hemisphere appear on a Direct Mercator chart?

[A] A straight line [B] A CURVED LINE curving towards the North Pole (convex to the Equator) [C] A curved line convex to the pole [D] A vertical line

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Because the Mercator projection stretches higher latitudes, a Great Circle curves northward, bowing away from the Equator towards the pole.

Q3: At what latitude on a Direct Mercator chart is the scale exactly DOUBLE the scale at the Equator?

[A] 30° [B] 60° ($\sec(60^\circ) = 1 / \cos(60^\circ) = 1 / 0.50 = 2.0$) [C] 45° [D] 90°

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Since scale expands as $\sec(\text{lat})$, at 60° latitude $\sec(60^\circ) = 2.0$, meaning the scale is exactly twice as large as at the Equator.